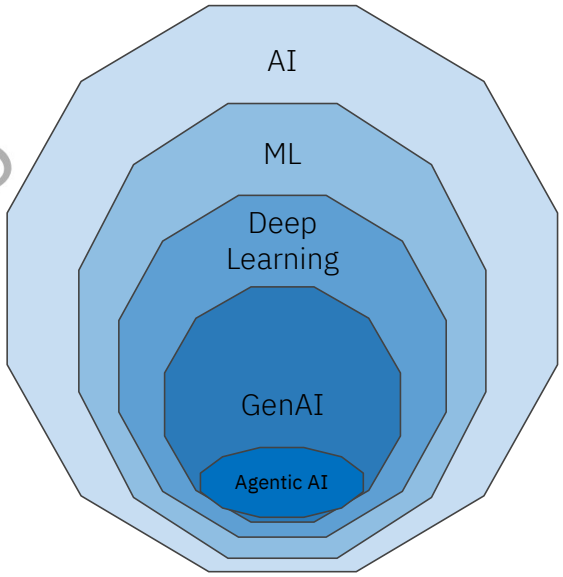
Introduction to AI

Week 1: AI Overview

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Data & AI Leader, IBM ASEAN

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Assistant

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Objectives

- Describe the basic concept of Artificial Intelligence (AI) and its importance in modern technology.
- Outline the history and key milestones of AI, from early beginnings to current advancements.
- Identify and explain key domains of AI, such as machine learning, natural language processing, computer vision, and robotics.

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AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation
W3: Search strategies

Machine Learning

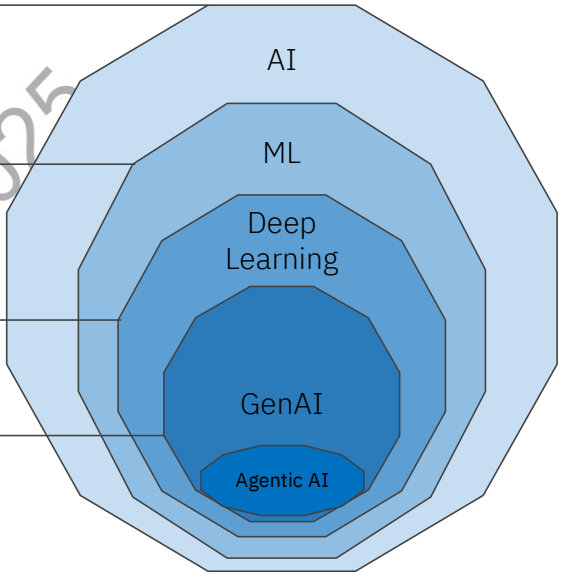
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron learning and neural network learning
W8: Deep learning

Generative AI

W9 + W10: Modern AI and development in real world applications (LLM & GenAI)



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Discussion

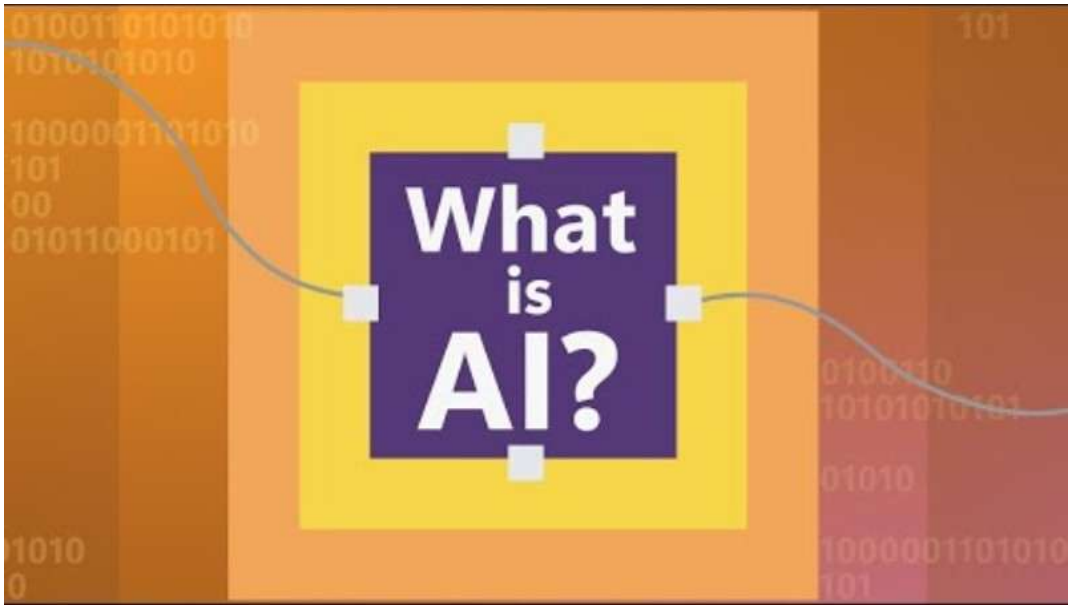
- TOP 3 keywords that come to your mind about AI?
- Which AI applications or tools have you used this week?
- What are your expectations and concerns — about AI, either this course or in general?

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What is AI?



https://www.youtube.com/watch?v=J4RqCSD--Dg

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AI Definition



The science and engineering of making intelligent machines

John McCarthy,
Father of AI, 1955



AI is the study of agents that perceive their environment and take actions that maximize their chance of successfully achieving their goals

Stuart Russell and Peter Norvig
In textbook "Artificial Intelligence: A Modern Approach"

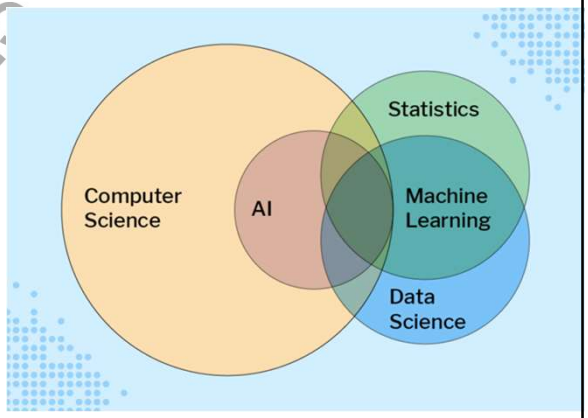
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What is AI?

- The study of computations that make it possible to perceive, reason and act
- A field of study that seeks to explain and emulate intelligent behaviour in terms of computational processes
- A branch of computer science that is concerned with the automation of intelligent behaviour
- Anything in Computing Science that we don't yet know how to do properly



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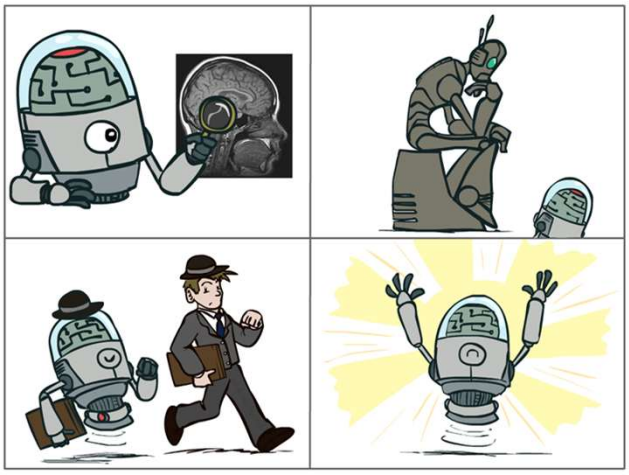
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AI and Humans

The science of making machines that:

Think like people

Duolingo models how people learn and forget vocabulary to schedule personalized practice.



Think rationally

A Sudoku solver uses formal logic and constraint solving to deduce the only valid solution.

Act like people

ChatGPT conducts natural conversations and answers questions in a human-like way.

Act rationally

YouTube and Spotify recommend content to maximize expected watch/listen time, choosing items that optimize that objective.

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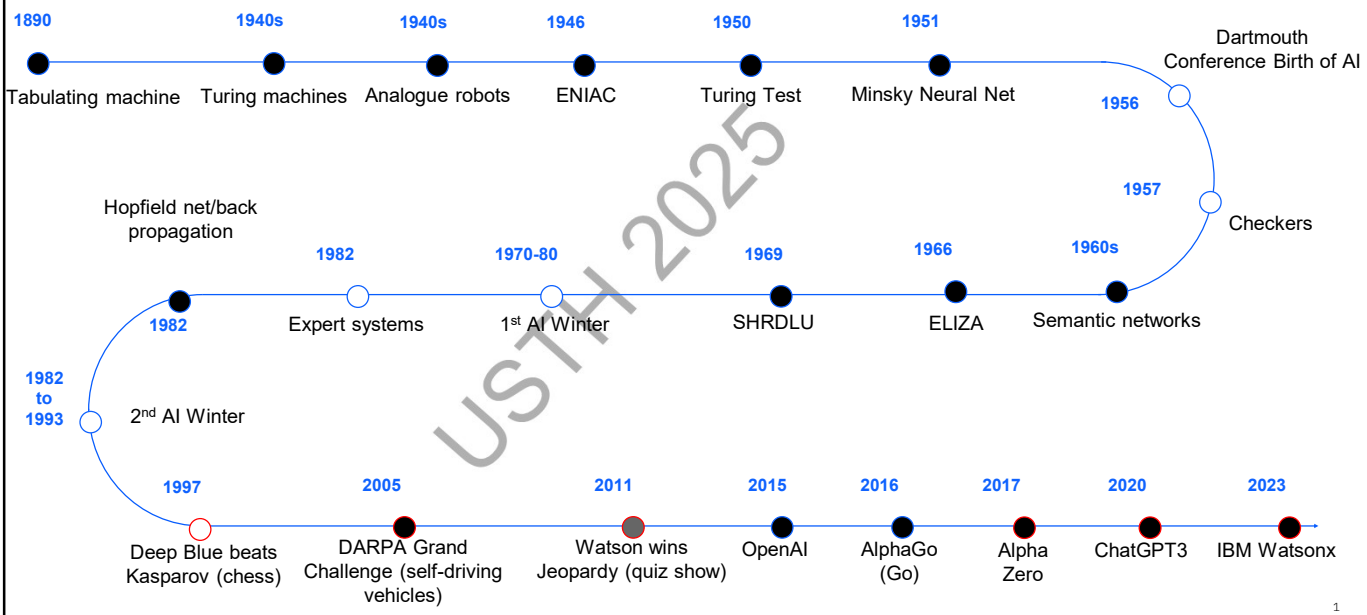
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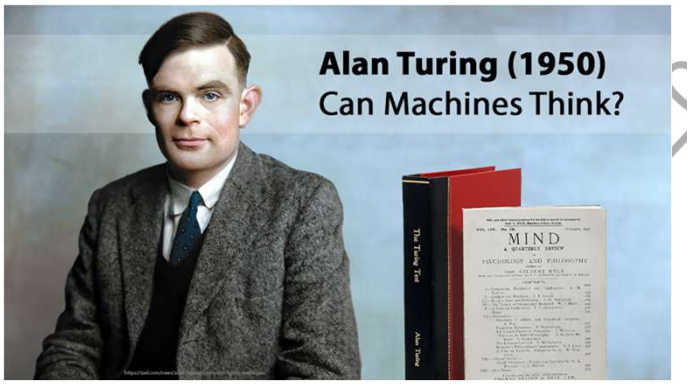
History of Computer Science

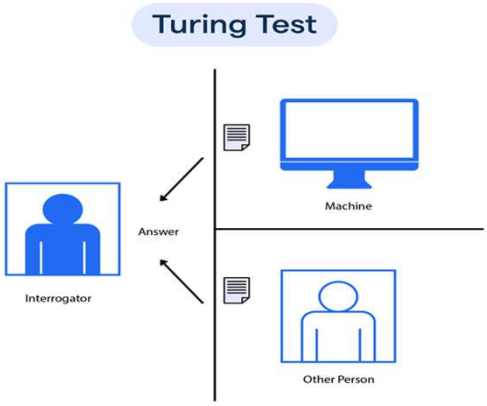


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1950 – "Can machines think?"





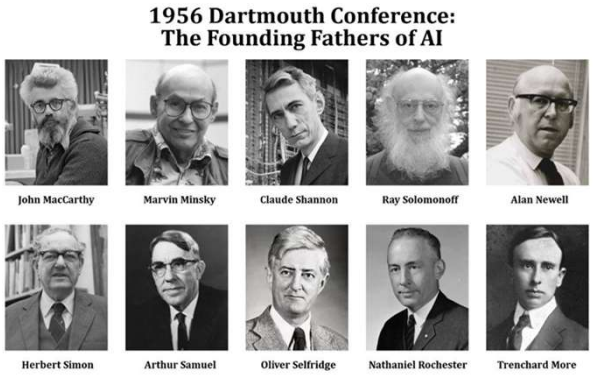
The machine is passed the test (to human-like intelligence) if interrogator can not distinguish the machine from human

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1956 – Birth of AI

- John McCarthy, an assistant professor of Mathematics at Dartmouth College, organized a group to explore if machines could simulate various types of human intelligence





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1997 – IBM Deep Blue

- 1st computer to beat a reigning champion with debates about AI surpassing human intelligence in strategy.
- Evaluated 200 million positions per second using specialized hardware and expert-tuned heuristics.
- Inspired future AI breakthroughs like AlphaGo and modern deep learning in games.



IBM's Deep Blue defeats world chess champion Garry Kasparov in a 6-game match.

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2006 – Deep Learning

- Advanced technology about data and computing
- Ignited the “Deep Learning” with Bengio and LeCun, shifting AI focus from hand-engineered features to learned hierarchical representations.
- Co-authored “Reducing the Dimensionality of Data with Neural Networks” – showed that deep autoencoders could discover low-dimensional manifolds in high-dimensional data (e.g., MNIST digits), outperforming PCA and kickstarting the deep learning revival.



Geoffrey Hinton
Nobel prize 2024
VinFuture prize 2024

- **Cảnh báo về AI**: VnExpress đưa tin về quyết định của Hinton rời Google để có thể nói lên quan ngại của mình về các mối nguy của AI mà không bị ràng buộc bởi lợi ích của công ty.
- **Giải thưởng**: Hinton là một trong những người nhận giải Turing năm 2018, giải thưởng được ví như giải Nobel trong lĩnh vực khoa học máy tính, cùng với Yoshua Bengio và Yann LeCun vì những đóng góp cho AI.

VnExpress
'Bố già' Geoffrey Hinton: Phát triển AI như nuôi hổ con
Apr 29, 2025 — Giáo sư Geoffrey Hinton, được mệnh danh là "Bố già AI", cho rằng AI có thể gây nguy hiểm cho nhân loại nếu "nuôi dạy" sai hướng.

VnExpress
'Bố già AI' cảnh báo gì về nguy cơ trí tuệ nhân tạo xóa sổ ...
Dec 28, 2024 — Nhà khoa học máy tính giành giải Nobel Vật lý 2024 Geoffrey Hinton cảnh báo có 10-20% khả năng AI sẽ hủy diệt nhân loại trong vòng ba thập ...

VnExpress
Chú nhân giải Nobel lo AI 'trỗi dậy'
Oct 9, 2024 — Geoffrey Hinton, chú nhân giải Nobel Vật lý 2024, được mệnh danh là cha đỡ đầu của AI, lo lắng về khả năng AI thông minh vượt con người.

VnExpress
'Bố già AI' bị bạn gái chia tay bằng ChatGPT
Sep 7, 2025 — Hinton, 78 tuổi, được mệnh danh là "Bố già AI" nhờ những đóng góp cho mạng neuron nhân tạo và được trao giải Nobel Vật lý năm ngoái. Ông thừa ...

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2011 – IBM Watson won Jeopardy

- IBM Watson defeats Human Champions on Jeopardy, earning $1m
- AI can answer complex natural language questions with 71% accuracy
- Pioneered Open Questions – combining linguistic parsing, hypothesis generation, evidence retrieval from massive corpora, and confidence-based ranking—shifting AI from narrow rule-based systems to probabilistic, evidence-driven reasoning over real-world language.
- Processed 200 Million Pages of Content in Real Time without internet access during the game



The first public demonstration of AI mastering unstructured human knowledge at scale.

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Quiz

- Can an AI model ever beat its own creator?

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2016 – Alpha Go

- AlphaGo—created by DeepMind—defeated world champion Lee Sedol 4–1 (Lee Sedol's level at 9-dan progressional)
- A feat previously thought impossible for machines due to the game's immense complexity
- AlphaGo's creators, including Demis Hassabis and David Silver (amateurs at level 1-3 dan), could not themselves play at that superhuman level



AI can surpass the abilities of the humans who built it

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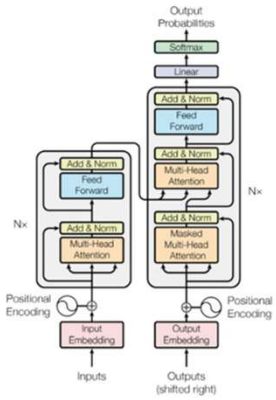
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2017 – Google Brain

- Paper “attention is all you need”
- Proposed a novel neural network based on self-attention mechanisms to enable parallel processing of sequences and achieve state-of-the-art results in machine translation with 8x faster training than prior RNN models.

Transformer

Attention Is All You Need



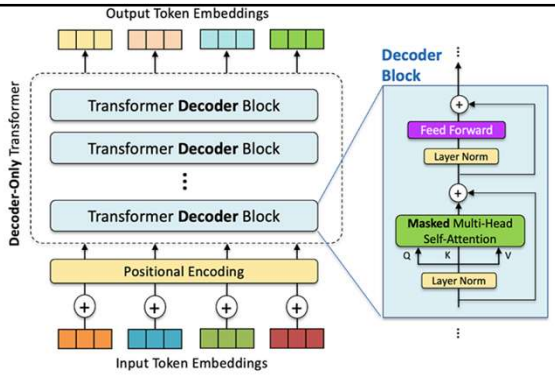
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2018 – Open AI

GPT



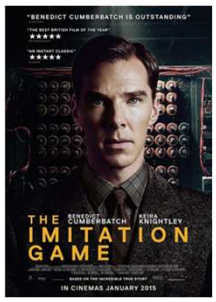
- 2018 GPT-1: Generative Pre-trained Transformer to improve language understanding
- 2019 GPT-2: Scaled to Human-Level Text Generation
- 2020 GPT-3: Demonstrated In-Context Learning
- 2023 GPT-4: Achieved Multimodal and Expert-Level Reasoning
- 2024-2025 GPT-4o & o1 Series: Real-Time Multimodality and Reasoning Chains

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Documentary



1940 Turing Machines



1997 - Deep Blue vs. Kasparov



2005 - Stanley vs. The DARPA Challenge



2016 - AlphaGo beats on one Go Master
2017- AlphaGo Zero vs. itself…



2015 – OpenAI
2020 – GPT-3
2022 – ChatGPT (GPT-3.5)
2023 – GPT-4

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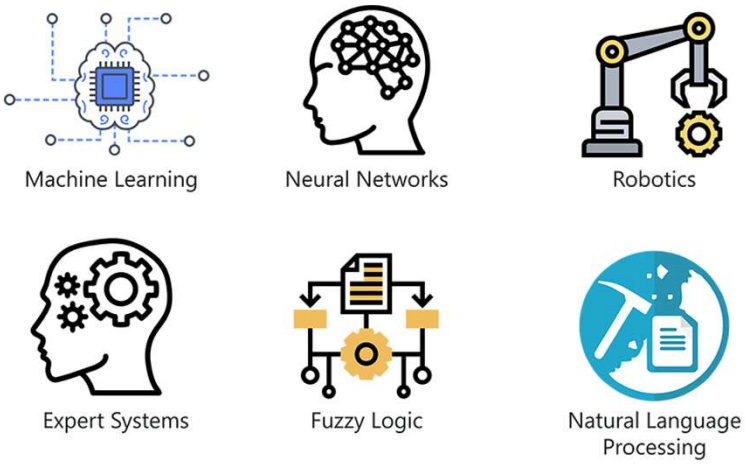
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Types of AI Domains



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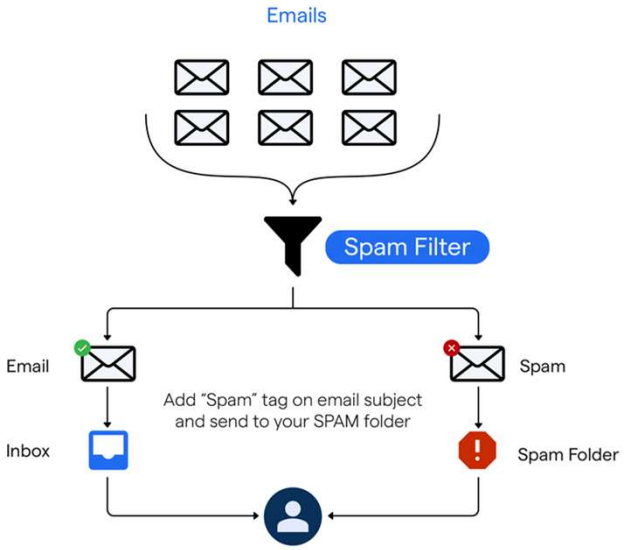
Machine Learning (ML)

Spam Filtering

Learns patterns from data: Models are trained on large datasets of labeled emails (“spam” vs “not spam”).

Extracts features: Looks at keywords, sender info, frequency of suspicious links.

Adapts over time: Improves accuracy as it sees new types of spam (phishing, promotions, etc.).



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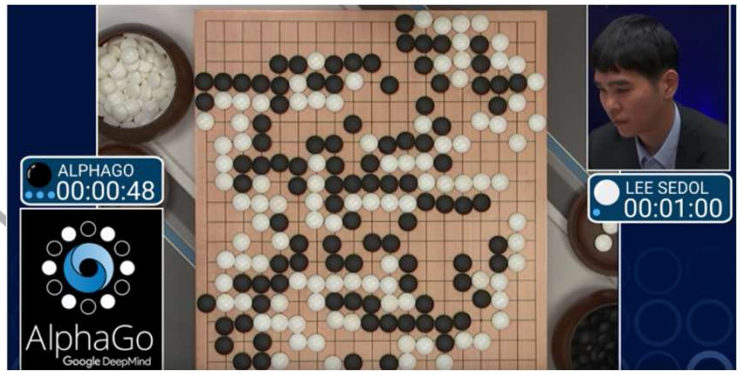
Deep Learning (DL)

AlphaGo (Google DeepMind)

Neural networks + search: Learned to evaluate Go board positions and plan winning moves.

Trained by self-play: Played millions of games against itself to improve beyond human knowledge.

Famous achievement: Defeated world champion Lee Sedol in 2016 — a milestone in AI history.



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Natural Language Processing (NLP)

Siri / Alexa

Speech recognition: Converts your spoken words into text using deep learning.

Language understanding: Interprets intent (e.g., “play music” or “set alarm”).

Response generation: Uses rules and ML to answer, control devices, or perform tasks.



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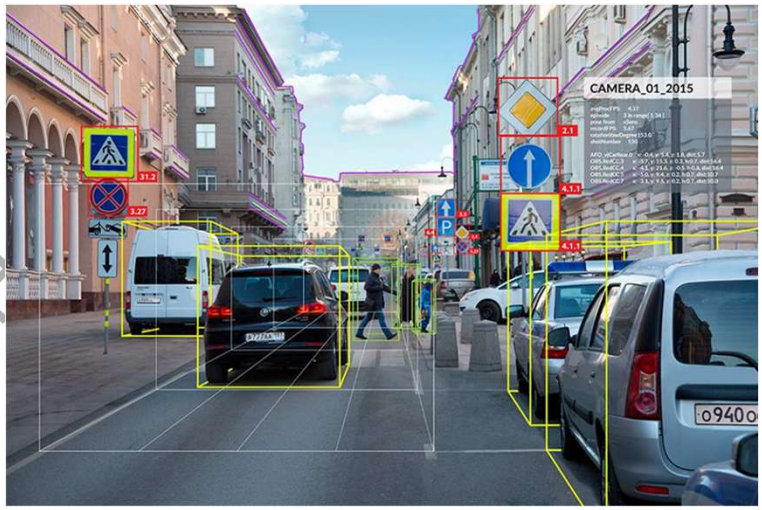
Computer Vision (CV)

Self-driving Cars

Perception of environment: Cameras, LiDAR, and radar feed into CV models to detect lanes, cars, pedestrians.

Object detection & segmentation: Identifies traffic lights, road signs, and obstacles in real time.

Integration with control: Outputs feed into planning and decision modules for safe driving.



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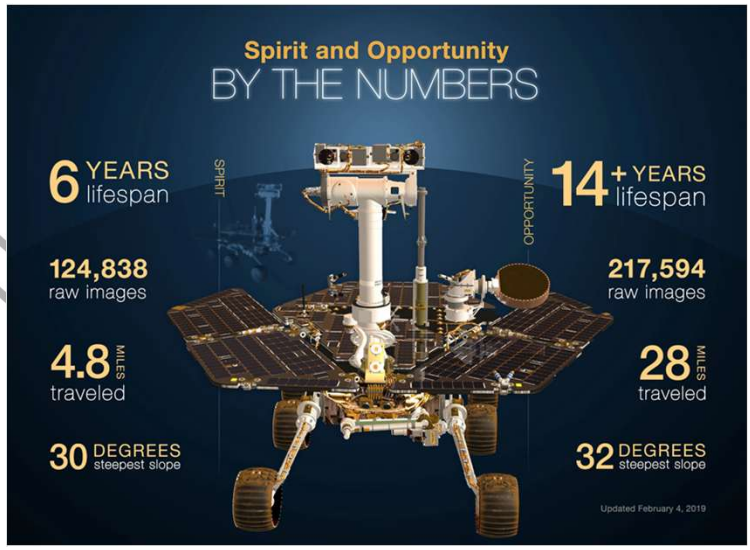
Planning

NASA Mars Rover

Autonomous scheduling: Plans tasks like driving, sampling, and charging within energy limits.

Search in state space: Considers terrain, obstacles, and distance to targets when plotting a route.

Handles uncertainty: Must act with limited communication (20 min delay to Earth).



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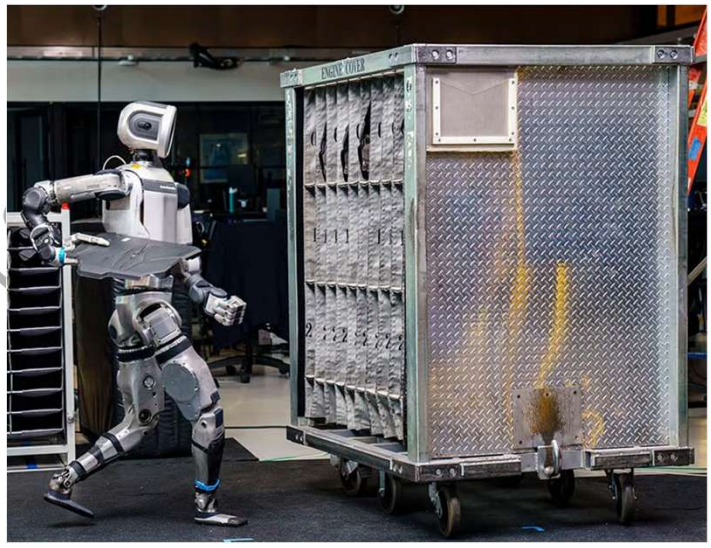
Robotics

Boston Dynamics Atlas

Perception & balance: Uses computer vision and sensors to navigate uneven terrain.

Planning motions: AI algorithms coordinate limbs for walking, running, or parkour.

Human-robot interaction: Demonstrates agility in dynamic environments, a step toward useful field robots.

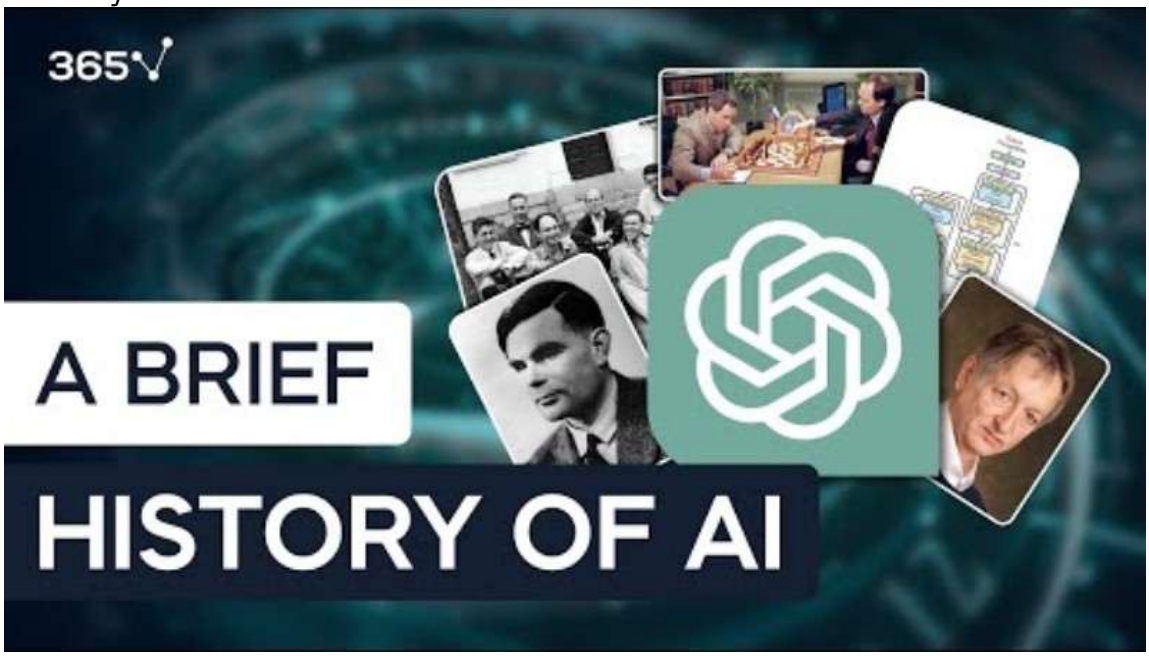


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Summary



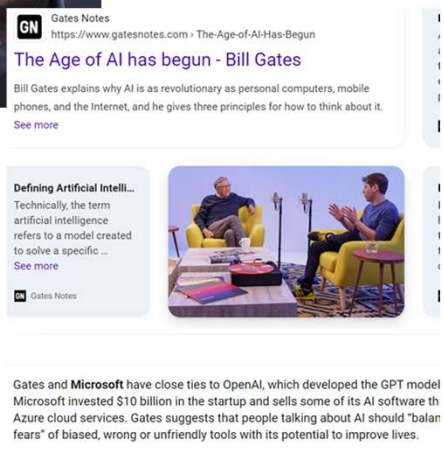
https://www.youtube.com/watch?v=eSj80Zr6TEE

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What's Next with AI?





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Thank you!

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